

National Tsing Hua University
11320IEEM 513600
Deep Learning and Industrial Applications
Homework 3

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Due on 2025/04/10.

Note: DO NOT exceed 3 pages.

1. (10 points) Download the MVTec Anomaly Detection Dataset from Kaggle ([here](#)). Select one type of product from the dataset. Document the following details about your dataset:
 - Number of defect classes: 7
 - Types of defect classes: bent wire, cable swap, cut inner insulation, cut outer insulation, missing cable, missing wire, poke insulation
 - Number of images used in your dataset: 139
 - Distribution of training and test data: 80%(109張) Train, 20%(30張) Test
 - Image dimensions: (channel: 3, height: 1024, width: 1024)
2. (30 points) Implement 4 different attempts to improve the model's performance trained on the dataset you choose in previous question. Ensure that at least one approach involves modifying the pre-trained model from TorchVision. Summarize the outcomes of each attempt, highlighting the best performing model and the key factors contributing to its success. You may also need to describe other hyperparameters you use in your experiment, like epochs, learning rate, and optimizer. (Approximately 150 words.)

Baseline: epochs = 75, learning rate = 1e-3, optimizer = Adam

1. Change Resize: 將照片的Resize從(32, 32)到(224, 224)
2. Change Batch Size: 接續2. 後，再將 Batch Size 從 32 改成8
3. Change lr&lr_scheduler: 接續3.，將learning rate從1e-3改成1e-2，並將learning rate scheduler從CosineAnnealingLR改成 ReduceLROnPlateau(factor = 0.5, patience = 5)
4. Using MobileNetV2: 接續4. 但將 Pre-train model 改成 MobileNetV2

Approach	Train Accuracy	Best Val Accuracy
Baseline	45.45%	53.33%

Change Resize	85.32%	83.33%
Change Batch Size	78.90%	86.67%
Change lr&lr_scheduler	89.00%	90.00%
Using MobileNetV2	80.73%	76.67%

表現最好的是第四個，但從前面將 Resize 改成224提供模型更多資訊時，準確度就有顯著的提升了，而因為資料料較小，所以也將Batch size降低，最後用 ReduceLROnPlateau 相對於原本的 CosineAnnealingLR 可以防止Learning rate過早下降。

3. (20 points) In real-world datasets, we often encounter long-tail distribution (or data imbalance). In MVTec AD dataset, you may observe that there are more images categorized under the 'Good' class compared to images for each defect class. (Approximately 150 words.)

(i) (5 points) Define what is 'long-tail distribution.'

資料集中在少數的類別，而多數的類別的資料量確很少，形成所謂的長尾。

(ii) (15 points) Identify and summarize a paper published after 2020 that proposes a solution to data imbalance. Explain how their method could be applied to our case.

提出 GRU-MACGANs 的模型，結合 GRU 與改良型生成對抗網路，根據用戶欺炸行為類型產生多模態的異常資料，提升少數類別樣本的多樣性。

雖然這篇論文是針對電力的時序資料，但是他提出的LOF和輔助分類器、Wasserstein loss + gradient penalty都可以在使用ACGAN生成圖片後，來提升樣本多樣性與穩定性，來平衡不同類別資料的數量。

Reference: Gao, B., Kong, X., Li, S., Chen, Y., Zhang, X., Liu, Z., & Lv, W. (2024). Enhancing anomaly detection accuracy and interpretability in low-quality and class imbalanced data: A comprehensive approach. *Applied Energy*, 353, 122157.

4. (20 points) The MVTec AD dataset's training set primarily consists of 'good' images, lacking examples of defects. Discuss strategies for developing an anomaly detection model under these conditions. (Approximately 100 words.)

參考論文中的GRU-MACGANs方法，利用改良的生成對抗網路生成多樣化的異常樣本，來增加少數類別的資料量。生成的樣本經分類器與異常檢測器驗證後再加入訓練集，提升分類器對異常的辨識能力，改善資料不平衡問題。

5. For the task of anomaly detection, it may be advantageous to employ more sophisticated computer vision techniques such as object detection or segmentation. This approach will aid in identifying defects within the images more accurately. Furthermore, there are numerous open-source models designed for general applications that can be utilized for this purpose, including YOLO-World ([website](#)) and SAM ([website](#)). (Approximately 150 words.)

- (i) (10 points) To leverage these powerful models and fine-tune them using our dataset, it is necessary to prepare specific types of datasets. What kind of data should be prepared for object detection and for segmentation.

對於物件偵測 (Object Detection)，需準備含有框出目標物的邊界框 (Bounding Box) 與標籤 (Labels) 的資料集；而對於分割 (Segmentation)，則需具備像素級別的標註 (Pixel-wise Annotations)，每個像素對應其對應的分類。

- (ii) (10 points) Why are these models suitable for fine-tuning for our custom dataset?

YOLO-World 和 SAM 這類模型具備強大的預訓練特性，能從大量通用資料學到豐富特徵。針對特定數據集進行微調時，可保留已有的特徵提取能力，同時能達到客製的目的。